Ref. No.: Ex/EE/PC/B/T/225/2026

B. E. ELECTRICAL ENGINEERING 2ND YEAR 2ND SEMESTER EXAMINATION, 2026

Subject: SEQUENTIAL SYSTEMS & MICROPROCESSOR Time: 3 Hours Full Marks: 100

Use a separate Answer-script for each Part

Part I (50 marks)

Question No. | **Question 1** is compulsory | Marks

Answer ***Any Two*** questions from the rest (2×20)

Q1 Answer ***any Two*** of the following: (2× 5 =10)

(a) With the help of a schematic diagram discuss the basic differences between a Combinational logic circuit and a Sequential logic circuit. 5

(b) A fictitious flip-flop with two inputs A and B is designed such that for AB= 00 and 11 the output becomes 0 and 1, respectively. For AB= 01, flip-flop retains previous output while output complements for AB= 10.
Determine the truth table and excitation table for this flip-flop. 5

(c) Explain, with a schematic diagram, how does the State Transition Diagram for a Moore Machine differ from that for a Mealy Machine? 5

(d) What is a Finite State Machine? Obtain State Transition Diagram of SR-Flip-Flop. 5

Q2 (a) All the flip-flops in the circuit shown in Figure Q2(a) are initially reset.

(i) Describe, with the help of state table, the operation of the circuit. 6

(ii) Draw the timing diagram for the circuit. 4

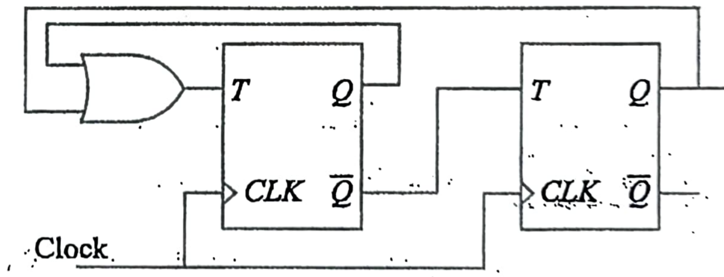
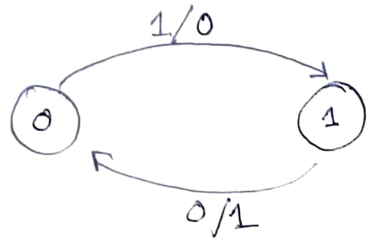


Figure Q2(a)

(b) (i) What is Excitation Table for a Flip-flop? How is it different from the Truth Table? 2+2

(ii) With the help of Excitation Table show how D-flip-flop can be converted to SR-flip-flop. 6



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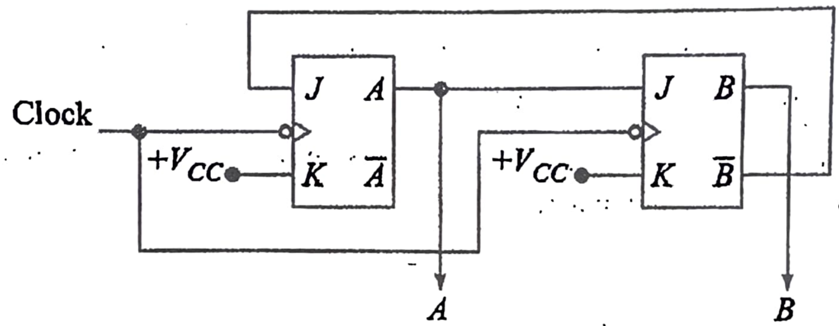
Q3 (a) (i) Justify that the circuit shown in Figure Q3(a) is a Mod-3 counter. 4

(ii) Show that if one connects the B output of the counter to the clock input of another J-K Flip-Flop, with both J and K of the 3rd Flip-Flop being connected to $+V_{CC}$, then the circuit becomes a Mod-6 counter. 6



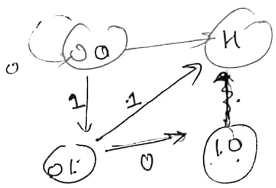


Figure Q3(a)

(b) Enumerate, with the help of schematic diagrams, different types of Shift Register configurations that are commercially available. 4

(c) With the help of a schematic diagram explain the functioning of a 4-bit parallel-in–serial-out shift right register realized using D flip-flops. 6

Q4 (a) (i) Discuss the major differences between Synchronous and Asynchronous Counters. 2

(ii) Design a MOD-6 asynchronous counter using T-Flip-Flops. ✓ 8

(b) Design a synchronous sequential circuit having the state transition diagram, as shown in Figure Q4(b), with the help of JK flip-flops. 10

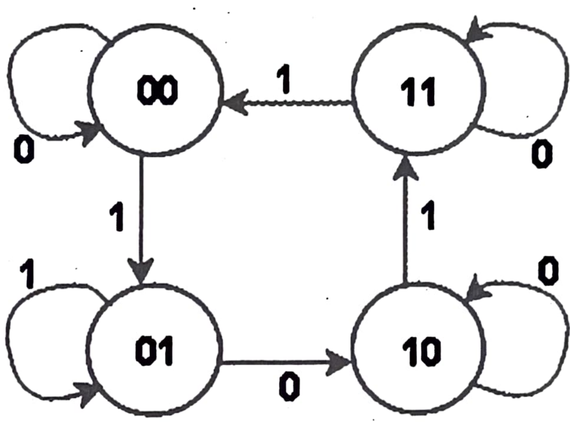


Figure Q4(b)

0 → 00 → 00 ✓
1 → 0 → 1 ✓
1 → 1 → 1 ✓
0 → 1 → 0
0 → 0 → 0
1 → 0 → 1 ✓
0 → 1 → 1 ✓
1 → 1 → 0

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Q5 (a) What is a State Transition diagram? Show how the State Transition Diagram for a Mealy Model as shown in Figure Q5(a) can be converted to the equivalent Moore Model diagrams. 2+2

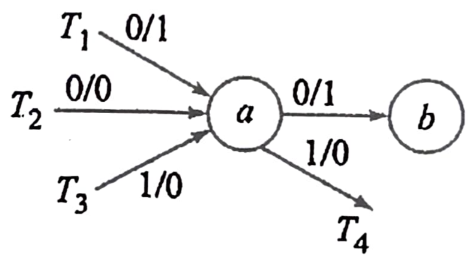

Figure Q5(a)

(b) Show how a Modulo-4 counter designed with JK flip-flops can generate a repeatative sequence of the binary word "1101" with minimum number of memory elements. 6

(c) The task is to design a synchronous logic control unit of a vending machine. The machine can take only two types of coins of denomination 1 and 2 in any order. It delivers only one product that is priced Rs. 3. On receiving Rs. 3 the product is delivered by asserting an output D = 1 which otherwise remains 0. If it gets Rs. 4 then product is delivered by asserting X and also a coin return mechanism is activated by output Y = 1 to return a Re. 1 coin.

There are two sensors to sense the denomination of the coins that give binary output as shown in the table.

I	*J*	**Coin**
0	X	No coin is inserted
1	0	One rupee Coin
1	1	Two rupees Coin

The clock speed is much higher than human response time, i.e. no two coins can be deposited in same clock cycle.

(i) Draw the ASM chart for a Mealy Model of the problem. 4

(ii) Derive the corresponding State Transition Diagram. 6

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B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER - 2026

SEQUENTIAL SYSTEMS & MICROPROCESSOR

PART-II(50 Marks)

Time: Three Hours   Use separate answer-script for each part   Full Marks: 100

Answer any THREE questions(2 marks reserved for neat and well-organized answer)

1. (a) Draw the functional pin diagram of 8085 microprocessor. (8)

   (b) With a functional block diagram and the corresponding timing diagram, discuss the Opcode Fetch cycle associated with the instruction MOV B,A. (8)

2. (a) Define addressing modes and explain the four types used in the 8085: Implied, Immediate, Direct, and Indirect. Provide one assembly language example for each. (8)

   (b) Derive the formula for calculating time delays in the 8085. Explain how a nested loop structure can be used to generate a delay of approximately 28.7 ms if the processor frequency is 2 MHz. (8)

3. (a) Describe the internal logic of 8085 interrupts. Explain how the SIM (Set Interrupt Mask) instruction interprets the accumulator bit pattern to enable or disable individual RST interrupts. (8)

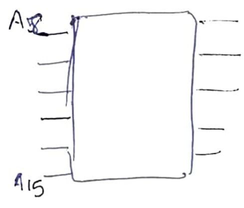
   (b) Explain the mechanism of the CALL and RET instructions. Describe how the Stack Pointer and the stack memory are used to store return addresses and preserve register contents using PUSH and POP. . (8)

4. (a) Explain what happens after the execution of the following instructions: (i) MVI M, data (ii) LDA address (iii) XTHL (iv)DAA (v)CMP M (vi)SPHL (3)

   (b) Debug the following 8085 programs and correct these with justification for each correction.

   (i)
   ```
         MVI C, FFH
   LOOP: INR C
         JNZ LOOP
         HLT
   ```

   (ii)
   ```
         LXI SP, FFFFH
         PUSH B
         PUSH D
         POP B
         POP D
         HLT
   ```



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(3+3)

   (c) Draw interfacing diagram for interfacing a memory IC(2K size) and I/O with 8085 CPU at address 2000H for memory and 07H for I/O. (7)

5. (a) Write assembly language program in 8085 to add two 8-bit data, the result being 16 bit. (8)

   (b) Write assembly language program in 8085 to convert 7FH to binary. (8)

0111 1111